MILITARY SPECIFICATION

# TRAINING AID: MOCKUP, KINGSBURY THRUST BEARING

This specification has been approved by the Naval Training Device Center, Department of the Navy.

## 1. SCOPE

1.1 This specification establishes the requirements for the production of a three dimensional mockup take apart model, which will serve as a classroom training aid to illustrate the mechanism, principles of operation, components arrangement, and nomenclature of a Kingsbury Thrust Bearing.

1.1.1 This item has been designated by the Naval Training Device Center, as Device 29HA6 for Navy procurement.

## 2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

| | |
|---|---|
| MIL-C-14366 | Nylon Twill, cloth coated |
| MIL-T-23991 | Training Devices, Military; General Specification for |

STANDARDS

Federal

| | |
|---|---|
| FED-STD-595 | Colors |

Military

| | |
|---|---|
| MIL-STD-129 | Marking for Shipment and Storage |
| MIL-STD-794 | Procedures for Packaging and Packing of parts and equipment |

DRAWINGS



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Naval Training Device Center (NAVTRADEVCEN)

| | |
|---|---|
| Code Identification No. 08013 Drawings No. 1 through 12 and document list | Device 29HA6, Kingsbury Thrust Bearing |

PHOTOGRAPHS

Naval Training Device Center (NAVTRADEVCEN)

| | |
|---|---|
| NP-40-890-8-55 through NP-40-893-8-55 | Device 29HA6, Kingsbury Thrust Bearing |

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer.)

3. REQUIREMENTS

3.1 Materials - Materials shall be as specified in 3.1.1 through 3.1.3.1 and as indicated on the drawings of 3.2. Materials not specified herein, nor on the drawings of 3.2, shall conform to the requirements of MIL-T-23991.

3.1.1 Metals - Metal parts shall be of a corrosion-resistant material, or of a material given a corrosion-resistant treatment or coating.

3.1.2 Wood - Wood used shall be of select grades of hardwood lumber, having good appearance and finishing qualities, suitable for either clear or opaque finishes. Wood shall be properly seasoned, free from checks, splits, and crossgrain or seasoning defects which may cause warping.

3.1.2.1 Plywood - Plywood shall have a minimum of five plies, or solid core hardwood, both surfaces to be of hardwood, all plies to be free of spaces or checks and joined by moisture resistant glue.

3.1.2.2 Glue - Glue shall not be adversely affected by moisture, lacquer, paint, migratory solvents of plastics used in components or, by the environmental conditions specified in 3.3.

3.1.3 Plastics - Plastics shall be durable and resistant to cracking, brittleness, discoloration, deformation and similar defects, upon aging or when subjected to environmental conditions specified in 3.3.

3.1.3.1 Plastic cementing - Contact surfaces of plastic pieces to be joined shall be machined or sanded. Wherever possible, a cushion surface shall be developed on one contact surface. Cement solvents shall be used as recommended by the manufacturer of the particular plastic used and tested by the contractor for each specific application to conform to the requirements of 3.3. Cemented joints shall be free of bubbles and form a strong homogeneous bond.

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3.2 Design - The training aid shall be a three-dimensional mockup, take apart model of a Kingsbury thrust bearing, mounted on a wood base, the design of which is depicted on drawings Code Identification No. 08013 and NAVTRADEVCEN photographs NP-40-890-8-55 through NP-40-893-8-55.

3.3 Reliability - The training aid shall be designed for a useful classroom life of 20 hours per week for 5 years. Reliability shall be a prime consideration in the design and manufacture of the training aid. The design shall be such as to result in reduced requirements for adjustments or alinement. The training aid shall be constructed so no fixed part shall become free and no moving part shall become loose or sluggish in operation. Degradation shall not be caused because of shipment, normal handling in classroom use or storage. The training aid shall be designed for storage and transportation when under ambient conditions of temperature ranging from minus 5 to plus 125 degrees Fahrenheit, relative humidity ranging up to ninety-five percent, and still be capable of meeting the specification requirements.

3.4 Construction - Construction of the training aid shall be in accordance with drawings of 3.2, the processes requirements of MIL-T-23991, and as specified in 3.5 through 3.9.

3.5 Dimensions and tolerances -

3.5.1 Dimensions - Dimensions not otherwise specified in the drawings of 3.2, shall be consistent with the functional design of the training aid and inherent qualities of the material being used.

3.5.2 Tolerances - Tolerances in construction, not otherwise specified in the drawings of 3.2, shall be consistent with functional design, inherent qualities of the material being used, and shall afford freedom from binding in structural and mechanical parts.

3.6 Details of components - The training aid shall consist of the following major components:

(a) Mockup

(b) Controls

(c) Base

(d) Accessories

(e) Dust cover

3.6.1 Mockup - The mockup of the Kingsbury thrust bearing shall be a full scale cutaway, take apart model of a Kingsbury thrust bearing with an 18.875 inch collar, its components in three dimensional form, and parts color coded. The components shall be stylized, but sufficiently detailed so that the training aid can be manually operated, disassembled and reassembled, and



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measurements taken of simulated bearing wear. The thrust unit shall be contained in a clear plastic shell with removable cover plate. All major components of the bearing shall be visible, including the upper and lower leveling plates, which shall be capable of being viewed through a sectionalized portion of one base ring, and the cover plate.

3.6.2 Controls - Parts shall be movable, manually actuated and rotated by means of a handle, to demonstrate how the Kingsbury thrust bearing and components function, their interaction and relationship.

3.6.2.1 Turning the handle shall provide rotation of the main rotor shaft which in turn shall actuate the bearing components.

3.6.3 Base - The mockup of the Kingsbury thrust bearing shall be mounted within a wooden frame simulating the housing, similar in shape and size to the operational Kingsbury reduction gear or turbine housings. The training aid mockup and housing shall be mounted on a wooden base. The base shall also support an attached box of accessories, and a bridge gage.

3.6.4 Accessories - The training aid shall include the following accessories:

(a) Bridge gage

(b) Micrometer depth gage

(c) Feeler gage.

3.6.4.1 The micrometer depth gage and feeler gage for measuring bearing wear shall be stored in a fitted wood case mounted on the base alongside the housing. Similarly, the bridge gage shall be held upright in place by stud bolts set in the wood base, on the opposite side of the training aid housing.

3.6.5 Dust Cover - A dust cover of cloth coated nylon twill type I per MIL-C-14366 or the equivalent shall be provided.

3.6.5.1 The dust cover shall be labeled with black lettering of one-half inch size to identify the training aid by title and number.

3.7 Finish and color -

3.7.1 Finish - Surfaces to be painted shall be properly sealed, primed, and then finished with a satin lacquer in accordance with the Preparation for Painting and Painting requirements of MIL-T-23991. Application of finishes shall be accomplished in such a manner as not to impair performance of the training aid.

3.7.1.1 All aluminum parts shall be anodized in accordance with MIL-A-8625.

3.7.2 Colors - The training aid components, interconnecting sections, and fittings, shall be color coded to distinguish structural, mechanical and operational functions as indicated on the drawings of 3.2.



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3.7.2.1 Colors shall be in accordance with FED-STD-595

3.8 Nameplates and product markings -

3.8.1 Nameplates and emblems - Nameplates and emblems shall be as specified by the procuring activity and in accordance with contract requirements. Nameplates and emblems shall be attached to the training aid in the location indicated on the photographs of 3.2.

3.8.2 Markings - All components of the training aid mockup shall be identified by the use of numerals and titles on component listing placards. These placards and all components shall be labeled and identified in accordance with the drawings and photographs of 3.2.

3.8.2.1 The placards shall be labeled with engraved lettering on plastic filled with a contrasting color. Plastic with nonreflective surface, similar to LAMACOID or equal, shall be used.

3.9 Workmanship - The training aid, including all parts and accessories, shall be constructed and finished in a thoroughly workmanlike manner. Particular attention shall be given to neatness and thoroughness and markings and indications of parts and assemblies, painting, and freedom of parts from burrs and sharp edges.

## 4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection - Inspections to be performed are classified as follows:

(a) In-process inspection

(b) Quality conformance inspection

4.3 In-process inspection - In-process inspection shall include such visual and mechanical examination, and testing of materials, subassemblies, parts and accessories (including purchased standard items) during the manufacturing process as may be necessary to assure conformance to all the requirements of this specification.

4.4 Quality conformance inspection - Quality conformance inspection shall consist of a preproduction inspection and inspection of production units.



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4.4.1 Preproduction inspection - A preproduction inspection shall be conducted on a preproduction unit and shall consist of all the examinations and tests specified in 4.5 and 4.6.

4.4.2 Production inspection - Production units shall be inspected to the extent necessary to maintain quality control and conformance to the Government-accepted preproduction unit.

4.5 Examinations - Examinations shall be performed to verify conformance to the following requirements:

(a) Materials

(b) Design

(c) Construction

(d) Dimensions and tolerances

(e) Details of components

(f) Finish and color

(g) Nameplates and product markings

(h) Workmanship

(i) Preparation for delivery.

4.6 Tests - Functional tests shall be conducted to determine compliance to the requirements of this specification.

## 5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging - Procedures for preservation and packaging shall be in accordance with MIL-STD-794. Level of preservation and packaging shall be as specified in the contract schedule.

5.3 Marking - Interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.

## 6. NOTES

6.1 Intended use - The training aid mockup is intended to be used by instructors as an aid in group training of Naval personnel, or as a self-training medium, in understanding the nomenclature of components, theory and principles of operation, as well as the steps to disassemble, determine bearing clearance, and reassemble a Kingsbury thrust bearing.

6.2 Ordering data - Procurement documents should specify the following:

(a) Title, number and date of this specification



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(b) Procuring activity designation (see 1.1.1)

(c) Nameplates and product markings (3.8)

(d) Selection of applicable levels of preservation, packaging and packing (see 5.1 and 5.2).

CUSTODIAN:
Navy-TD

PREPARING ACTIVITY
Navy-TD
Project No. 6910-N418



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DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-T-29026A(td) Training Aid: Mockup, Kingsbury Thrust Bearing.

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

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